III Coarse Cohomology

3.0. Reminder

- A coarse structure on a set X is a family of binary relations $\mathcal{E} \subseteq \mathcal{P}(X \times X)$, called entourages or controlled sets, which satisfies:

  i) $E_1 \subseteq E_2 \in \mathcal{E} \Rightarrow E_1 \in \mathcal{E}$.
  ii) $E_1, E_2 \in \mathcal{E} \Rightarrow E_1 \cup E_2 \in \mathcal{E}$
  } Ideal structure

  iii) $\Delta \in \mathcal{E}$.
  iv) $E_1, E_2 \in \mathcal{E} \Rightarrow E_2 \circ E_1 \in \mathcal{E}$.
  v) $E \in \mathcal{E} \Rightarrow E^{-1} \in \mathcal{E}$
  } Groupoid structure.

  $X \times X$
  $(y,z)(x,y) = (x,z)$

  Then $(X, \mathcal{E})$ is a coarse space.

- Examples:

  i) (X,d) metric space $\rightsquigarrow$ $\mathcal{E}_d = \left\{ E \subseteq X \times X \;\middle|\; \sup_{(x_1,x_2) \in E} d(x_1,x_2) < \infty \right\}$

  (Bdel) metric coarse structure

  ii) (X,d) metric space:

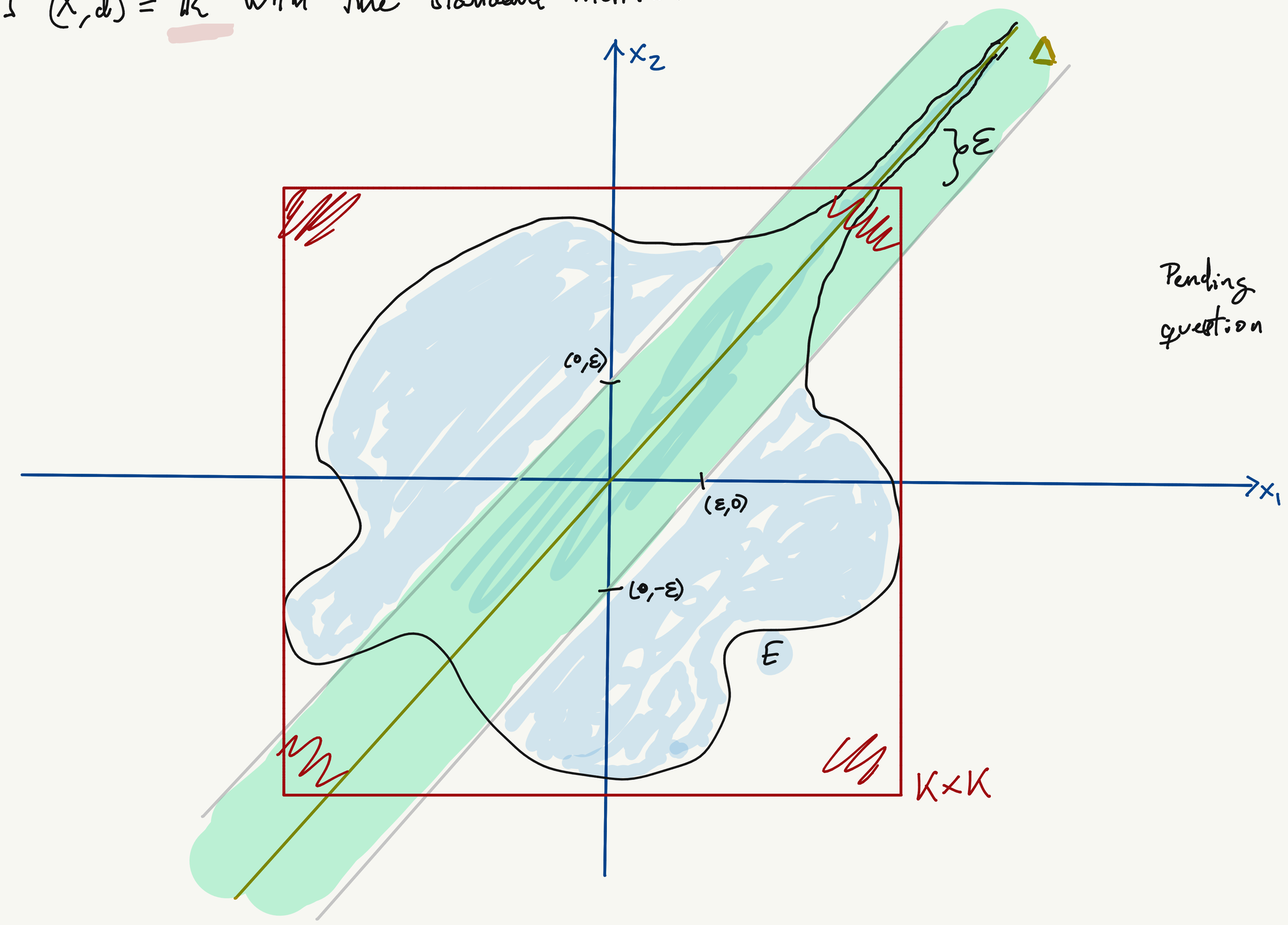
  $\rightsquigarrow$ $\mathcal{E}_{Co} = \left\{ E \subseteq X \times X \;\middle|\; \forall \varepsilon > 0 \; \exists K \subseteq X \text{ cpt s.t. } d|_{E - K \times K} < \varepsilon \right\}$

  Co coarse structure

  $\rightarrow$ Example in $X = (0,1)$:

Here $(0,1)^2$ is metrically bdeel, but not every $E \subseteq X \times X$ is Co-controlled.

-] $(X,d) = \mathbb{R}$ with the standard metric.



$$T_\varepsilon = \{(x_1,x_2) \in \mathbb{R}^2 \mid d(x_1,x_2) < \varepsilon\} = \{x_1 - \varepsilon < x_2 < x_1 + \varepsilon\}$$

E is controlled but not bded.

iii) $i: X \hookrightarrow \overline{X}$ compactification of a paracompact locally compact Hausdorff space.

$\partial X = \overline{X} - X$ "corona" of the compactification.

$$\mathcal{E}_\iota = \{E \subseteq X\times X \mid \overline{E} \setminus X\times X \subseteq \Delta\partial X\} =$$

$$= \{E \subseteq X\times X \mid \overline{E} \cap (\partial X \times \overline{X} \cup \overline{X}\times\partial X) \subseteq \Delta\partial X\}$$

3.1. Coarse connectedness

- Let $(X, \mathcal{E})$ be a coarse space. For $x, y \in X$,

$x \sim y$ "x and y are in the same coarsely connected component"

$\overset{\text{def}}{\iff} \exists\, E \in \mathcal{E}$ s.t. $(x,y) \in E$

$\iff \{(x,y)\} \in \mathcal{E}$

$F = \{(x,y)\} \subseteq X \times X$;
$F \subseteq E \in \mathcal{E}$
$\Rightarrow F \in \mathcal{E}$

$E := \{(x,y)\}$

$\sim$ is an equivalence relation.

The coarsely connected components of $(X, \mathcal{E})$ are the equivalence classes under $\sim$.

$(X, \mathcal{E})$ is "coarsely connected" if there is at most one coarsely conn. component.

- Example:
  -) An extended metric space $(X, d: X \times X \to [0, +\infty])$ endowed w/ the bded metric coarse str., satisfies:

    $x \sim y \iff (x,y) < \infty$.

    $x \sim y \iff \{(x,y)\} \in \mathcal{E}$

    (X,d) is coarsely conn. iff $d(x,y) < \infty \ \forall x,y \in X$, i.e. (X,d) is a metric space.

    $\overline{\mathbb{R}} = [-\infty, +\infty]$
    $d(-\infty, x) = \infty \quad \forall x \neq -\infty$
    $d(+\infty, x) = \infty \quad \forall x \neq +\infty$.

    $(\overline{\mathbb{R}}, d: \overline{\mathbb{R}} \times \overline{\mathbb{R}} \to [0,\infty])$ has three coarsely connected components:

    $\{-\infty\}, \ \mathbb{R}, \ \{+\infty\}$.

  -) To a graph $\Gamma = (V, E)$ it corresponds an extended metric space (V, d), where

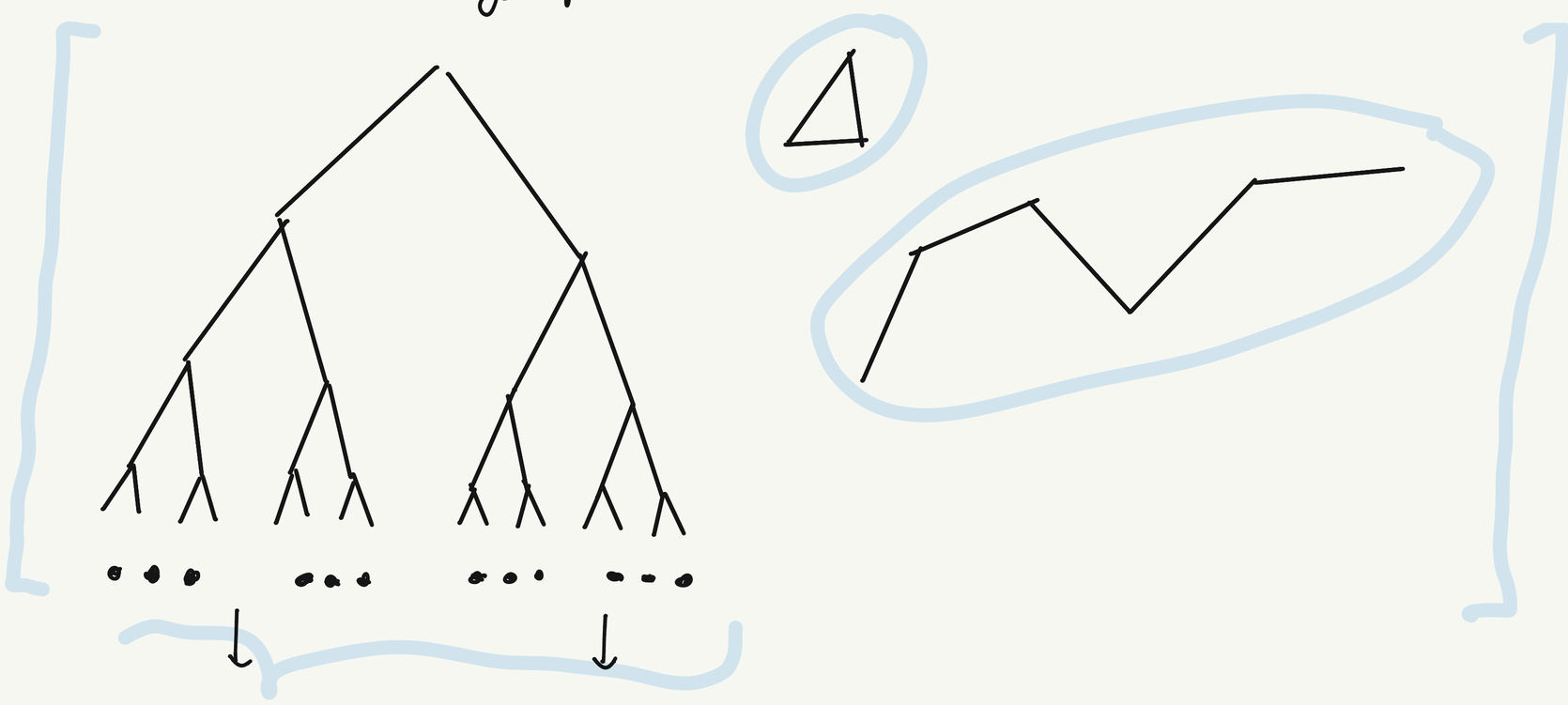
    $d(x,y) = \inf \{ \text{len}(\gamma) \mid \gamma \text{ path from } x \text{ to } y \}$ in $[0, \infty]$

    If x, y are not in the same conn. component of Γ, then

    $d(x,y) = \inf \emptyset = +\infty$.

    (V,d) is coarsely connected $\iff$ d being a metric
    $\iff \Gamma = (V, E)$ is connected.

E.g.
Consider the graph $\Gamma = (V, E)$



Γ has three coarsely conn. components, one is unbded and two bded.

- One restrict attention to coarsely connected spaces and then argue independently on each component for non coarsely connected spaces.

3.2. Closeness of maps and bounded sets

- **Def**: Let X be any set and $(Y, \mathcal{E}_Y)$ a coarse space. Two maps $f, g : X \rightrightarrows Y$ are "close", $f \sim g$, iff $\exists E \in \mathcal{E}_Y$ s.t. $(f(x), g(x)) \in E$ $\forall x \in X$.

$$X \underset{g}{\overset{f}{\rightrightarrows}} Y \; ; \quad f \times g : X \times X \to Y \times Y, \quad (x_1, x_2) \mapsto (f(x_1), g(x_2))$$

Equivalently,

$$f \sim g \iff (f \times g)(\Delta_X) = \{(f(x), g(x)) \mid x \in X\} \text{ is controlled } (\subset E \in \mathcal{E}_Y)$$

- **Example**: Assume (Y, d) is a metric space and endow it with the bded metric coarse structure

$$\mathcal{E}_d = \{E \subseteq Y \times Y \mid \exists r > 0 \text{ s.t. } d(x_1, x_2) \leq r \;\; \forall (x_1, x_2) \in E\}$$

$$X \underset{g}{\overset{f}{\rightrightarrows}} (Y, d) \; ; \quad f \sim g \iff \exists r > 0 \text{ s.t. } d(f(x), g(x)) \leq r \;\; \forall x \in X.$$

- $\sim$ closeness of maps is an equivalence relation on the set of maps $X \to Y$.

- **Remark**: If (X, d) is a metric space and $\mathcal{E}_d$ its bded metric coarse structure, then, for $B \subseteq X$,

$$B \text{ is bded} \iff \operatorname{diam} B < \infty \; ; \iff B \times B \in \mathcal{E}_d.$$

$$\operatorname{diam} B = \sup_{(x_1, x_2) \in B \times B} d(x_1, x_2) < \infty$$

- **Def**: Let $(X, \mathcal{E})$ be a coarse space. $B \subseteq X$ is said to be "bounded" if it satisfies one of the following equivalent conditions:

  i) $B \times B \in \mathcal{E}$ ($B \times B$ is controlled)

  ii) $\exists x \in X$ s.t. $\overbrace{Bx}^{}$ is controlled (For $\mathcal{E}_d$ if (X,d) is a metric sp., $\exists r > 0$ s.t. $d(y,x) \leq r \ \forall y \in B$; $B \subseteq \overline{B}(x,r)$)

  iii) $\exists E \in \mathcal{E}$ and $x \in X$ s.t. $B = E[\{x\}] = \{y \in X \mid (x,y) \in E\}$

  iv) $i: B \hookrightarrow X$ is close to a constant map.

- **Examples**:

  i) (X,d) bded metric structure $\mathcal{E}_d$.

  $B \subseteq X$ bded wrt. $\mathcal{E}_d \iff B$ is d-bounded

  ii) (X,d) metric space. $\rightarrow \mathcal{E}_{Co}$ coarse structure;

  $\mathcal{E}_{Co} = \{ E \subseteq X \times X \mid \forall \varepsilon > 0 \ \exists K \subseteq X$ cpt st. $d|_{E - K \times K} < \varepsilon \}$

  Let $B \subseteq X$.

  [B rel. cpt $\overset{(a)}{\Longrightarrow}$ B is bded wrt. $\mathcal{E}_{Co}$ $\overset{(b)}{\Longrightarrow}$ d-bounded]

  (a) $\forall \varepsilon > 0$, choose $K = \overline{B}$; $\underbrace{B \times B} - K \times K = \emptyset$

  $B \times B$ controlled $\rightarrow$ B bded wrt. $\mathcal{E}_{Co}$.

  (b) Choose $\varepsilon > 0$. Since $B \times B \in \mathcal{E}_{Co}$, $\exists K \subseteq X$ cpt. st. $d|_{B \times B - K \times K} < \varepsilon$.

  $B \times B - K \times K = (B - K) \times B \cup B \times (B - K)$

  If $B - K = \emptyset$, $B \subseteq K$ and K is cpt, then B is bded.
  otherwise, choose $x \in B - K$, and then for $y \in B$,
  $(x,y) \in (B - K) \times B$, $d(x,y) < \varepsilon$; $B \subseteq B(x, \varepsilon)$, B d-bded

  Conclusion: Iff (X,d) is a proper metric space,

  B bded wrt. $\mathcal{E}_{Co} \iff B$ d-bded $\iff B$ rel. cpt.

iii) $i : X \hookrightarrow \overline{X}$ compactification. $\underbrace{\partial X}_{\text{"corona"}} = \overline{X} - X$

$$\mathcal{E}_i = \{ E \subseteq X \times X \mid \overline{E} - X \times X \subseteq \Delta_{\partial X} \}$$

$\overline{E} \subseteq \overline{X} \times \overline{X}$

Let $B \subseteq X$. Then

B bded $\iff B \times B \in \mathcal{E}_i \iff \underbrace{\overline{(B \times B)} \cap (\partial X \times \overline{X} \cup \overline{X} \times \partial X)}_{\overline{B} \times \overline{B}} \subseteq \Delta_{\partial X}$

$$(\overline{B} \cap \partial X) \times \overline{B} \cup \overline{B} \times (\overline{B} \cap \partial X) \subseteq \Delta_{\partial X}$$

If $\overline{B} \cap \partial X = \emptyset$, then the condition is satisfied.

so B is bded.

If $\overline{B} \cap \partial X \neq \emptyset$, in particular $\overline{B} \neq \emptyset$, and so $B \neq \emptyset$.

We choose $w \in \overline{B} \cap \partial X$ and $y \in B$. Then

$(w, y) \in (\overline{B} \cap \partial X) \times \overline{B} \quad \not\subseteq \Delta_{\partial X}$

so B is not bded.

Then:

B is bded $\iff \overline{B} \cap \partial X = \emptyset \iff \overline{B}^{\overline{X}} \subseteq X$

3.3. Maps between coarse spaces

* **Def**: Let $f:(X,\mathcal{E}_X)\to(Y,\mathcal{E}_Y)$ be a map between coarse spaces.

  ·) f is "controlled" if $\forall E\in\mathcal{E}_X$, $\underbrace{(f\times f)(E)}_{\{(f(x_1),f(x_2))\mid (x_1,x_2)\in E\}}\in\mathcal{E}_Y$.

  f is "controlled" if $\forall E\in\mathcal{E}_X\ \exists F\in\mathcal{E}_Y$ s.t.

  $$(x_1,x_2)\in E \implies (f(x_1),f(x_2))\in F.$$

  ·) f is "proper" if $\forall B\subseteq Y$ bded, $f^{-1}(B)\subseteq X$ is bded.

  ·) f is a "coarse map" if it is controlled and proper.

The category "Coarse" has:

·) Objects: Coarse spaces $(X,\mathcal{E}_X)$

·) $\mathrm{Coarse}(X,Y)=\{$ coarse maps $X\to Y\}$

| Composition of coarse maps is coarse.

Coarse

- Weak equivalences (coarse equivalence)
- Category w/ weak equivalences
- Fib, Cofib?

$(X,\mathcal{E})$

$(X,\mathcal{E}_X)\longrightarrow(Y,\mathcal{E}_Y)$

Top

- Weak eq.
- Fib.
- Cofib.

(X,τ)

$(X,\tau_X)\longrightarrow(Y,\tau_Y)$

$X\underset{g}{\overset{f}{\rightrightarrows}}Y$; $f\simeq g$.

Coarse$/_{\simeq}$

$f_1\simeq f_2$, $g_1\simeq g_2\Rightarrow g_1f_1\simeq g_2f_2$

ob: $(X,\mathcal{E})$ coarse spaces

$\hom(X,Y)=\{[f]\mid f:X\xrightarrow[\text{coarse}]{}Y\}$

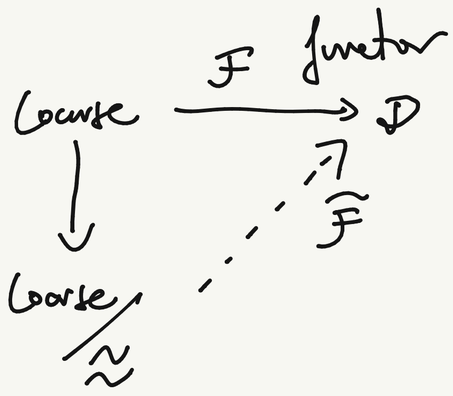
↑ eq. class up to closeness.

Top$/_{\simeq}$

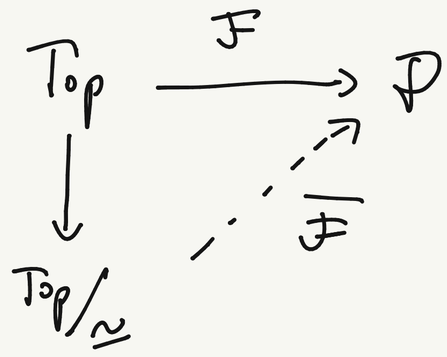
ob: (X,τ) top. spaces

$\hom(X,Y)=\{[f]\mid f:X\xrightarrow[\text{continuous}]{}Y\}$

↑ eq. class under $\simeq$



Coarse invariants



Homotopy invariants

If F : Coarse $\to \mathcal{D}$ is a coarse invariant, then

$$X \overset{\sim}{ce} Y \quad \Longrightarrow \quad F(X) \cong F(Y).$$

(isomorphic in Coarse/$\overset{\sim}{ce}$)

- $f : (X, \mathcal{E}) \longrightarrow (Y, \overbrace{\mathcal{F}(Y \times Y)}^{\text{maximal coarse structure}})$

f is controlled $\forall\, \mathcal{E}$ coarse str on X

f is proper iff $(X, \mathcal{E})$ is bded, iff $\mathcal{E} = \mathcal{F}(X \times X)$.

Control: Obj $(X, \mathcal{E})$ coarse spaces

hom: controlled maps.

$$\Big(f_i : \underset{\text{set}}{X} \longrightarrow (Y_i, \mathcal{E}_i)\Big)_{i \in I} \quad \text{U-source}$$

$\mathcal{E}$ ini coarse structure on X; the largest coarse structure on X such that f_i is controlled $\forall i \in I$

$(X,d) \longrightarrow$ Bded metric structure $\mathcal{E}_d$

ii) $\Rightarrow$ Connectedness

iii) $\Rightarrow$ Bdedness

$B \subseteq X$ is bounded wrt. $\mathcal{E}_d \iff B \times B \in \mathcal{E}_d \iff \exists r > 0$ s.t.
$d(x_1, x_2) \leq r \quad \forall (x_1, x_2) \in B \times B \iff \exists r \,/\, \operatorname{diam} B \leq r \iff B$ d-bded

i) Coarse connectedness vs path-conn. vs connectedness.

(X,d) is coarsely connected iff $d(x,y) < \infty \quad \forall x,y \in X$

Path-connected $\Rightarrow$ coarsely conn. (Heine–Cantor Theorem).

$\not\Leftarrow$ E.g. G f.g. groups:

$(\mathbb{Z}, d)$ is coarsely conn but it is not path-conn.

$\operatorname{Cay}(G,S) \rightsquigarrow (G, d_S)$ not path-conn., but coarsely equivalent to the geometric realization of $\operatorname{Cay}(G,S)$, which is path-conn.

$\Rightarrow$ Question: Given a metric space X (which is then coarsely conn.), is there a path-conn. metric space Y s.t. $X \overset{\sim}{c.e} Y$?

$X = \{ z^n \mid n \in \mathbb{N} \} \subseteq \mathbb{R}$ w/ the inherited metric,

X is not coarsely equivalent to a path-conn. metric space.
because X is not r-connected for any $r > 0$.
(Y is "r-connected" if $\forall x,y \in Y \ \exists\, x_0 = x, x_1, \dots, x_n = y \in Y$
such that $d(x_i, x_{i+1}) \leq r \quad \forall i = 0, \dots, n-1$)

